

# MASSINISSA MEROUANI

Ph.D. Candidate at New York University

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## RESEARCH INTERESTS

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**Broad Interests:** Applied Machine Learning, Computer Systems, High-Performance Computing

**Focus Areas:** ML for Systems Optimization, AI-Driven Code Optimization, Efficient Deep Learning Systems, Generative AI for Automated Performance Engineering.

## EDUCATION

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**New York University - NYU Tandon School of Engineering**

**Sept. 2022 - Present**

*Ph.D. in Computer Science (GPA: 4.0/4.0)*

*New York, USA*

- Advisor: Professor Riyadh Baghdadi.
- Honors: Recipient of the NYUAD Global Ph.D. Fellowship.
- Expected Graduation: Fall 2026.

**École Nationale Supérieure d'Informatique**

**Sept. 2015 - Dec. 2020**

*Master's Degree in Computer Systems*

*Algiers, Algeria*

- Joint B.S. & M.S. 5-year Integrated Program.
- Graduation Thesis: A Deep Learning Based Cost Model for Automatic Code Optimization in Tiramisu.

## EXPERIENCE

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**New York University Abu Dhabi (NYUAD)**

**Sept. 2022 - Present**

*Graduate Research Assistant*

*Abu Dhabi, UAE*

- Conducting doctoral research on AI-driven code optimization within the Modern Compilers Lab.
- Designed end-to-end pipelines for automatic program scheduling, utilizing techniques ranging from supervised learning to LLM-based reasoning agents.
- Mentored undergraduate and master's students on systems and compiler optimization projects.

**New York University Abu Dhabi (NYUAD)**

**Jan. 2021 - Aug. 2022**

*Research Engineer*

*Abu Dhabi, UAE*

- Led engineering efforts to integrate ML optimization techniques into the Tiramisu compiler.
- Designed a distributed data collection framework to profile thousands of code schedules, generating large-scale labeled datasets to address data scarcity in automatic optimization.
- Architected and optimized deep learning models for code performance prediction, contributing to multiple research projects within the group.

**Massachusetts Institute of Technology (MIT)**

**Aug. 2019 - July 2020**

*Research Intern (CSAIL - COMMIT Group)*

*Cambridge, USA (Remote)*

- Developed the first LSTM-based cost model for the Tiramisu compiler, formulating the scheduling as a ranking problem; this approach significantly reduced search time and won the **Outstanding Paper Award** at MLSys 2021.

## SELECTED PUBLICATIONS

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**A Deep Learning Based Cost Model for Automatic Code Optimization**

**Mar. 2021**

*4th Conference on Machine Learning and Systems 2021 (MLSys 2021) [AR: 24%]*

R. Baghdadi, M. Merouani, H. Leghettas, et al.

- **Outstanding Paper Award**
- Selected for oral presentation at the **NeurIPS 2020 Machine Learning for Systems** workshop.
- Invited presentation at **MLSH'21** (Int. Workshop on Machine Learning for Software Hardware Co-Design).

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| <b>A Deep Learning Guided Exploration of Affine Unimodular Loop Transformations</b><br><i>12th Int. Workshop on Polyhedral Compilation Techniques (IMPACT 2022)</i><br>M. Merouani, K. Boudaoud, N. Aouadj, et al.                  | <b>June 2022</b>    |
| <b>A Deep Learning Model for Loop Interchange</b><br><i>32nd Int. Conference on Compiler Construction (CC 2023)</i><br>L. Mezdour, K. Kadem, M. Merouani, et al.                                                                    | <b>Feb. 2023</b>    |
| <b>LOOPer: A Learned Automatic Code Optimizer For Polyhedral Compilers</b><br><i>34th Int. Conference on Parallel Architectures and Compilation Techniques (PACT 2025) [AR: 28%]</i><br>M. Merouani, A. Boudaoud, N. Aouadj, et al. | <b>Nov. 2025</b>    |
| <b>Agentic Auto-Scheduling: A Study of LLM-Guided Loop Optimization</b><br><i>34th Int. Conference on Parallel Architectures and Compilation Techniques (PACT 2025) [AR: 28%]</i><br>M. Merouani, I. Kara Bernou, and R. Baghdadi.  | <b>Nov. 2025</b>    |
| <b>LOOPerSet: A Large-Scale Dataset for Data-Driven Compiler Optimization</b><br><i>25th Int. Symposium on Code Generation and Optimization (CGO 2027)</i><br>M. Merouani, A. Boudaoud, and R. Baghdadi.                            | <b>To Appear</b>    |
| <b>Fine-Tuning Large Language Models for Polyhedral Auto-Scheduling</b><br>• Distills a teacher LLM via SFT, then refines with GRPO to directly optimize runtime.                                                                   | <b>Under Review</b> |
| <b>Generative Schedule Prediction for Polyhedral Auto-Scheduling</b><br>• Replaces search with one-shot schedule prediction. Trains a model via Imitation Learning and Reinforced Self-Training.                                    | <b>Under Review</b> |

## AWARDS AND HONORS

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- **Outstanding Paper Award**, Conference on Machine Learning and Systems (MLSys) 2021
- **NYUAD Global Ph.D. Fellowship**, New York University, 2022 - Present
- **1st Place Rank National, Top 7.5% Worldwide**, Google HashCode Qualification Round, 2021
- **Top 6% Individual Rank**, AI Hack Tunisia, 2019

## TECHNICAL SKILLS

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|-----------------|-------------------------------------------------------------------------------------------------------------|
| <b>Core</b>     | Python, C/C++, PyTorch, OpenMP, Polyhedral Model, Distributed data & training pipelines, Slurm, Docker, Git |
| <b>Familiar</b> | CUDA, JAX, MLIR, MPI, Spark, Kubernetes                                                                     |

## SERVICE & TEACHING

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| <b>Teaching Assistant - Machine Learning</b><br><i>New York University Abu Dhabi</i><br>• Prepared and led lab/tutorial sessions, guiding students through practical ML exercises and theoretical reviews. | <b>Spring &amp; Fall 2025</b>     |
| <b>Invited Speaker</b><br><i>MLSH Workshop (ML for Software Hardware Co-Design)</i><br>• Presented lab research as an invited speaker at MLSH '21 and '23 editions.                                        | <b>Sept. 2021 &amp; Oct. 2023</b> |
| <b>Co-Organizer</b><br><i>GDG Algiers (Google Developers Group)</i><br>• Led organization of 12+ events and delivered Tech Talks on Docker and Kubernetes.                                                 | <b>2018 - 2020</b>                |
| <b>Research Mentor</b><br><i>New York University Abu Dhabi</i><br>• Supervised undergraduate and Master's student research projects on ML for compilers.                                                   | <b>Sept. 2021- Present</b>        |
| <b>Service for Conferences and Journals</b><br>• CC 2024 Artifact Evaluation Committee; Journal Reviewer (TACO, ParCo).                                                                                    |                                   |